Exo2 :

3) Détermination des armatures des éléments de la passerelle en BA (ELS)

$\overline{\sigma}_{bc} = 0{,}6 \cdot f_{c28} = 15$ MPa.

Fissuration préjudiciable → $\overline{\sigma}_{st} = \min\left\{\frac{2}{3} f_e ; 110\sqrt{\eta f_{tj}}\right\}$.

$\eta = 1{,}6$, Barre HA

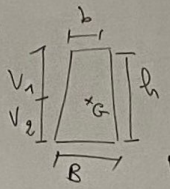
$\rightsquigarrow \overline{\sigma}_{st} = \min\{266{,}66 ; 201{,}633\} = 201{,}633$ MPa.

a) Armature de la dalle en console

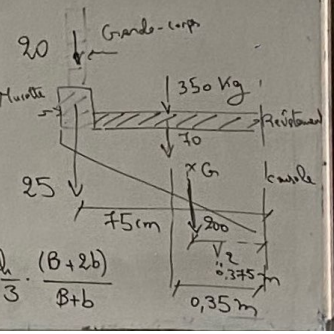
Evaluation des moments renversant à une console :

Grande Corps (d'un seul côté) $= 20 \cdot 0{,}75 = 15$ kg.m

Muretta $= 25 \times 0{,}75 = 18{,}75$ kg.m




$V_2 = \frac{h_t}{3} \cdot \frac{(B+2b)}{B+b}$




Console $= 200 \cdot (0{,}375) = 75$ kg.m

Revêtement $= 70 \cdot (0{,}35) = 24{,}50$ kg.m

Surcharge $= 350 \cdot (0{,}35) = 122{,}5$ kg.m

$M_{ELS} = 255{,}75$ kg.m.

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$$\mu = \frac{M_{ELS}}{b \cdot d^2 \cdot \overline{\sigma_s}} = \frac{255{,}75 \cdot 10 \cdot 10^2}{100 \cdot (9)^2 \cdot 201{,}633 \cdot 10^2}$$

$$\rightsquigarrow \mu = 0{,}00156$$

$$y_i = y_1 + (y_2 - y_1) \cdot \frac{(\mu_i - \mu_1)}{(\mu_2 - \mu_1)}$$

$$\beta_1 = 0{,}934 + (0{,}933 - 0{,}934) \cdot \frac{(0{,}00156 - 0{,}00152)}{(0{,}00157 - 0{,}00152)}$$

$$K_1 = 60{,}76 + (59{,}63 - 60{,}76) \cdot \frac{(0{,}00156 - 0{,}00152)}{(0{,}00157 - 0{,}00152)}$$

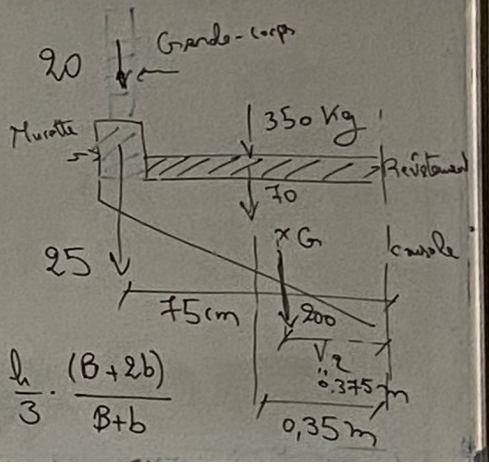
$\mu = 0{,}00156 \rightarrow K_1 = 59{,}856$ et $\beta_1 = 0{,}9332$.

$$\sigma_{bc} = \frac{\overline{\sigma_{st}}}{K_1} = \frac{201{,}633}{59{,}856} = 3{,}368 \text{ Mpa} < \overline{\sigma_{bc}} = 15 \text{ Mpa}$$

$\rightarrow$ (S.S.A)

$$A_{st}^{ELS} = \frac{M^{ELS}}{\beta_1 \cdot d \cdot \overline{\sigma_{st}}} = \frac{255{,}75 \cdot 10 \cdot 10^2}{0{,}9332 \cdot 9 \cdot 201{,}633 \cdot 10^2} = 1{,}51 \text{ cm}^2$$

La section d'Armature Trouvé à l'ELS est inférieure à celle adoptée à l'ELU.




$$V_2 = \frac{h}{3} \cdot \frac{(B + 2b)}{B + b}$$

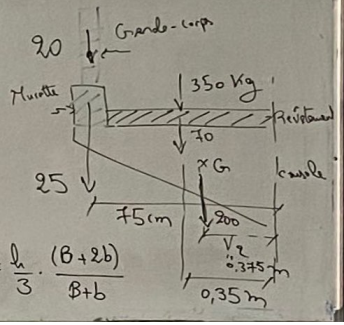
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$M = 0,00156 \rightarrow K_1 = 59,856$ et $B_1 = 0,9332$.

$\sigma_{bc} = \frac{\overline{\sigma_{st}}}{K_1} = \frac{201,633}{59,856} = 3,368\ Mpa < \overline{\sigma_{bc}} = 15\ Mpa$ $\rightarrow$ (S.S.A)

$A_{st}^{ELS} = \frac{M^{ELS}}{B_1 . d . \overline{\sigma_{st}}} = \frac{255,75 . 10 . 10^2}{0,9332 . 9 . 201,633 . 10^2} = 1,51\ cm^2$

La section d'Armature trouvé à l'ELS est inférieure à celle adoptée à l'ELU.

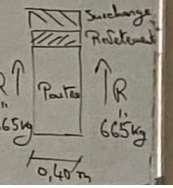



$V_2 = \frac{h}{3} . \frac{(B+2b)}{B+b}$

## b) Armature de la poutre

Evaluation des charges /ml :

- Réaction des consoles : $2 \times 665 = 1330$ kg /ml
- Poids propre de la poutre $\rightarrow 1000$ kg /ml
- Revêtement sur la poutre : $100 \times 24 = 40$ kg /ml
- Surcharge sur la poutre $= 500 \times 0,40 = 200$ kg /ml

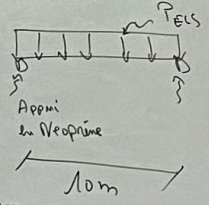



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$$P_{ELS} = (1330 + 1000 + 40) + (200) =$$

$$\rightsquigarrow P_{ELS} = 2570 \text{ kg/ml}$$

**Moment Maximal**




$$M^{max} = \frac{p^{ELS} \cdot l^2}{8}$$

$$M^{max} = \frac{2570 \cdot (10)^2}{8} = 32125 \text{ kg.m}$$

Le moment équilibré par la Table de compression :

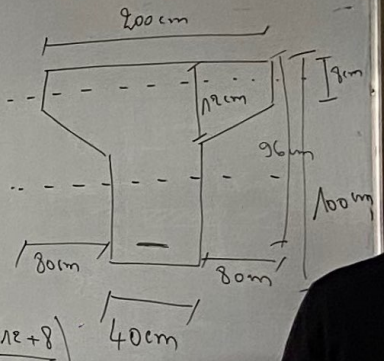
$$M_{tab} = f_{bu} \cdot b \cdot h_0 \cdot \left(d - \frac{h_0}{2}\right)$$

$$M_{tab} = 14{,}2 \cdot 10^5 \cdot 2 \cdot 0{,}10 \cdot \left(0{,}96 - \frac{0{,}10}{2}\right) = 258440 \text{ kg.m}$$

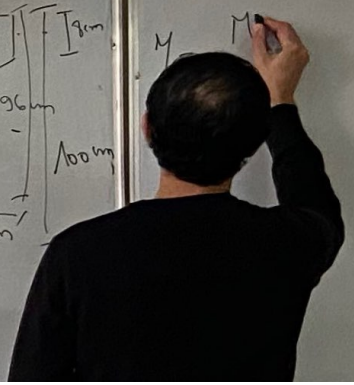
$M^{max}_{ELS} = 32125 \text{ kg.m} < M^{tab} = 258440 \text{ kg.m}$

→ l'axe neutre tombe dans la table de compression.

Le calcul se fera en considérant une section Rectangulaire (200 x 100).




$$h_0 = \left(\frac{12+8}{2}\right)$$




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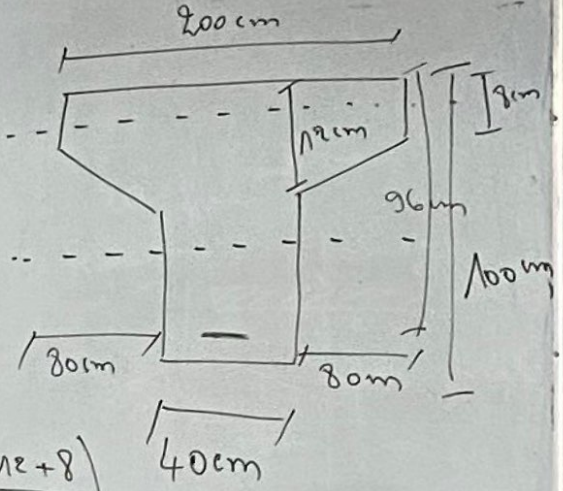
quilibré par la Table de compression:

$b.h_0.\left(d-\frac{h_0}{2}\right)$

$2.10^5.2.0,10.\left(0,96-\frac{0,10}{2}\right)=258440\ Kg.m$

$125\ kg.m < M^{tab}=258440\ kg.m$

a Tourbe dans la table de compression. $b \times h$
na en conticlénant une Section Rectangulaire $(200 \times 100)$.




$h_0=\left(\frac{12+8}{2}\right)$

$$\mu=\frac{M_{ELS}^{max}}{b.d^2.\bar{\sigma}_*}=\frac{32125.10.10^2}{200.(96)^2.201,533.10^2}$$

$$\rightsquigarrow \mu=0,00864$$

$$y_i=y_1+(y_2-y_1).\frac{(\mu_i-\mu_1)}{(\mu_2-\mu_1)}$$

$$B_1=0,950+(0,949-0,950).\frac{(0,000864-0,00084)}{(0,00087-0,00084)}$$

$$K_1=85+(83-85).\frac{(0,000864-0,00084)}{(0,00087-0,00084)}$$

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$\mu = 0{,}00864 \rightarrow K_1 = 83{,}4$ et $\beta_1 = 0{,}9492$

$\sigma_{bc} = \dfrac{\overline{\sigma}_{st}}{K_1} = \dfrac{201{,}633}{83{,}4} = 2{,}417$ Mpa

$\sigma_{bc} = 2{,}417$ Mpa $< \overline{\sigma}_{bc} = 15$ Mpa (S.S.A)

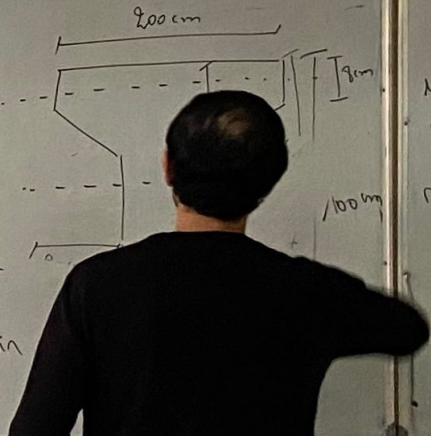
$A_{st} = \dfrac{M^{max}_{ELS}}{\beta_1 . d . \overline{\sigma}_{st}} = \dfrac{32125.10.10^2}{0{,}9492.96.201{,}633.10^2}$

$A^{ELS}_{st} = 17{,}48\ cm^2$

$A_{serv} = 17{,}48\ cm^2 > A_{ultim} = 15{,}70\ cm^2$

Le ferraillage longitudinale adopté : $A_{st} = 5HA25 = 19{,}64\ cm^2$

Il s'agit d'une poutre de grande hauteur, Alors il faut prévoir des armatures de peau constitués de 4HA10 par face




$\mu = \dfrac{M^{max}_{ELS}}{b . d^2 . \overline{\sigma}_{st}} = \dfrac{32125.10.10^2}{200.(96)^2.201{,}633.10^2}$

$\rightsquigarrow \mu = 0{,}00864$

$y_i = y_1 + (y_2 - y_1) . \dfrac{(\mu_i - \mu_1)}{(\mu_2 - \mu_1)}$

$\beta_1 = 0{,}950 + (0{,}949 - 0{,}950) . \dfrac{(0{,}00864 - 0{,}00084)}{(0{,}00087 - 0{,}00084)}$

$K_1 = 85 + (83 - 85) \dfrac{(0{,}00864 - 0{,}00084)}{(0{,}00087 - 0{,}00084)}$

Scanned with CamScanner

$\mu = 0,000864 \rightarrow K_1 = 83,4$ et $\beta_1 = 0,9492$

$$\sigma_{bc} = \frac{\overline{\sigma}_{st}}{K_1} = \frac{201,633}{83,4} = 2,417 \text{ Mpa}$$

$\sigma_{bc} = 2,417 \text{ Mpa} < \overline{\sigma}_{bc} = 15 \text{ Mpa}$ (S.S.A)

$$A_{st} = \frac{M_{ELS}^{max}}{\beta_1 . d . \overline{\sigma}_{st}} = \frac{32125.10.10^2}{0,9492.96.201,633.10^5}$$

$A_{st}^{ELS} = 17,48 \text{ cm}^2$

$A_{serv} = 17,48 \text{ cm}^2 > A_{ultim} = 15,70 \text{ cm}^2$

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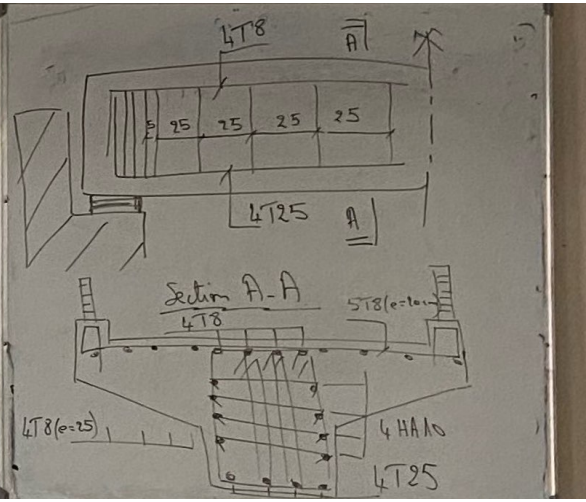
$$A_{st} = \frac{M_{ELS}^{max}}{\beta_1 . d . \overline{\sigma}_{st}} = \frac{32125.10.10^2}{0{,}9492.96.201{,}633.10^5}$$

$A_{st}^{ELS} = 17{,}48 \text{ cm}^2$

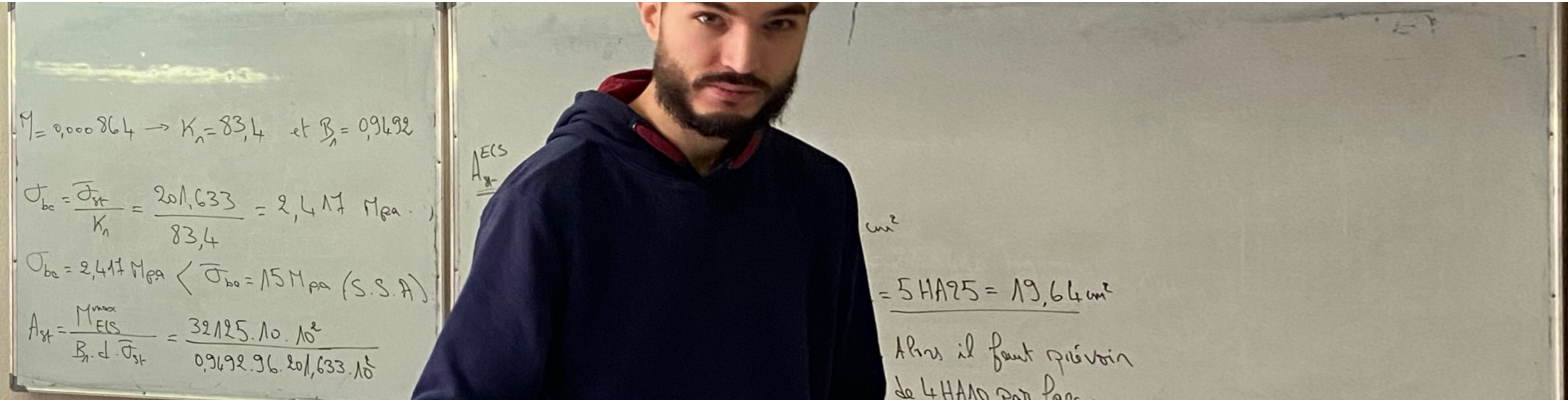
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$A_{st} = \frac{M^{max}_{ELS}}{\beta_1 . d . \bar{\sigma}_{st}} = \frac{32125.10.10^2}{0,9492.96.201,633.10^5}$
$A^{ELS}_{st}$
$cm^2$
$= 5HA25 = 19,64 cm^2$
Alors il faut prévoir